RISHII KUMAR SINGH

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github.com/RishiiGamer2201 | linkedin.com/in/rishiikumarsingh | Portfolio

SUMMARY

Environmental Engineering undergraduate at DTU doing AI research at AIMS Lab, DTU and UMTRI, University of Michigan, after an AI/ML internship with the Indian Navy. Builds Python tools, retrieval systems and vision pipelines.

EDUCATION

Delhi Technological University
B.Tech, Environmental Engineering | First-year CGPA: 8.35/10 *2025 - 2029 (expected)*

EXPERIENCE

Research Intern Jun 2026 - Present
AIMS Lab, DTU | Prof. Dinesh K. Vishwakarma
- Submitted a knowledge-graph completion paper, now under review at ECIR 2027.
- Leading restoration and Jetson deployment for the lab's MaCVi 2027 challenge team (WACV 2027 workshop).

Remote Research Collaborator Sep 2026 - Present
UMTRI, University of Michigan | Dr. Arpan Kusari
- Leading the next version of the MatPredict benchmark for material segmentation and inverse rendering.
- Reproduced the published results; built self-tested evaluation tooling with bootstrap confidence intervals.

AI/ML Intern Jun 2026 - Jul 2026
WESEE, Indian Navy | Ministry of Defence, Government of India
- Fine-tuned computer-vision models for satellite-image object detection and segmentation using Python and PyTorch.
- Reviewed data duplicates and split leakage, and supported active learning to improve evaluation-data quality.

AI & ML Intern Oct 2025 - Dec 2025
Artha Research & Intelligence Lab | Remote
- Preprocessed 488 facility records across 12 districts using Python and pandas for healthcare-accessibility analysis.
- Mapped facilities, settlements and road closures with Folium to visualize access context in Himachal Pradesh.

RESEARCH PROJECTS

- **CVPR 2027 project:** leading real-world data collection for a vision dataset and benchmark.
- **Learn2Design 2026:** solo NeurIPS 2026 Competition Track entry, optimizing designs in JAX; results pending.

PROJECTS

Nyaya Navigator | *Legal-information prototype* | GitHub
- Built a React/FastAPI legal assistant with local Gemma models and hybrid retrieval over 6,845 law-text chunks.
- Added fact confirmation and citation checks, withholding answers whose claims were flagged as unsupported.

Sherpa | *Local-model developer assistant* | GitHub | PyPI
- Built a Python CLI (Click, Rich, local GGUF models) that explains command output and answers code questions.
- Published 5 sherpa-dev releases through v0.1.4 on PyPI, distributing the assistant as an installable CLI.

MERGED OPEN-SOURCE CONTRIBUTIONS

- **Heliox-OS:** Contributed 10 PRs spanning plugin UI, WebSocket token streaming and model-loading race guards.
- **Upstream testing:** NumPy warning fixes in scikit-bio #2556; differential tests in scikit-verify #32.

TECHNICAL SKILLS

Languages & Web: Python, C/C++, JavaScript, TypeScript, SQL, FastAPI, Flask, React, PostgreSQL
ML & Tools: PyTorch, JAX, scikit-learn, NumPy, pandas, OpenCV, MediaPipe, ONNX, Git, pytest, Docker

ACHIEVEMENTS AND LEADERSHIP

- **Team hackathons:** Won Green Tag, BITS APOGEE 2026; 3rd place, BITCAMP, AIMS-DTU x TinyFish 2026.
- **GSSoC 2026:** Ranked 1,662 of 47,951 contributors (top 3.5%); 13 merged PRs across 3 projects.
- **AIMS-DTU:** Co-Head of the university's AI student society, Sep 2026 - Present.